Database Systems (CPH) – Fall 2025
Lecturer: Tiantian Liu
Exam Date: Jan 13, 2026

AALBORG UNIVERSITY
DENMARK

Preamble

Please read the following instructions carefully:

- This exam has seven questions and 14 pages. Make sure your copy contains all exercises and all questions. The following table summarizes the questions in this exam and their weights.

Question	**Points**
1. Relational Algebra	11
2. SQL	20
3. ER Modeling	16
4. Design Theory	14
5. Physical Design	12
6. Query Processing and Optimization	12
7. Transactions, Concurrency Control, and Recovery	15

- Read the text of each exercise carefully before solving it.
- If you get stuck on an exercise, proceed with another one and come back later.
- If your solution is based on an assumption that you have made, please add a comment for clarification.
- Aids are allowed during the exam, i.e., you can consult aids such as books, notes, etc.
- You may use materials on Moodle, but NO other online resources or help are allowed.
- As a reference, we recommend the course material we endorsed (book, slides, exercises, etc.) as it complies with the notation we introduced in the course, which is the same one used in the exam.
- It is NOT allowed to seek help from or consult others (human beings, artificial intelligence, etc.) during the exam.
- Good luck!

1. Relational Algebra (11 points)

Answer the following questions (Choose **only one** option):

1.1 (2 points) Which statement about a natural join is correct?

A. It always produces a Cartesian product
B. It joins two relations using all attributes with the same name
C. It requires an explicit join condition
D. It keeps duplicate join attributes in the result

1.2 Consider the following relations:

employee(*eid, name, age, dept*)
project(*pid, title, budget*)
works_on(*eid* → *employee, pid* → *project, hours*)

where the primary keys are underlined and the foreign keys are denoted by arrows.

1.2.1 (3 points) Which relational algebra expression returns the names of employees older than 35?

A. $\sigma_{age > 35}$ (Π_{name} (*employee*))
B. $\Pi_{name, age}$ ($\sigma_{age > 35}$ (*employee*))
C. Π_{name} ($\sigma_{age > 35}$ (*employee*))
D. σ_{name} (age > 35 (*employee*))

1.2.2 (3 points) Which expression finds all employees who work on at least one project, returning *eid* and *name*?

A. $\Pi_{eid, name}$ (*employee* × *works_on*)
B. $\Pi_{eid, name}$ (*works_on*)
C. $\Pi_{eid, name}$ ($\sigma_{hours} > 0$ (*employee*))
D. $\Pi_{eid, name}$ (*employee* ⋈ *works_on*)

1.2.3 (3 points) Which expression finds employees who do not work on any project, returning *eid*?

A. Π_{eid} (*employee* ⋈ *works_on*)
B. Π_{eid} (*employee*) – Π_{eid} (*works_on*)
C. Π_{eid} (*employee* × *works_on*)
D. Π_{eid} (σ_{pid} IS NULL (*works_on*))

2. SQL (20 points)

2.1 Evaluate whether the following statements about SQL are true or false.

2.1.1 (2 points) In SQL, a condition in the HAVING clause may refer to an aggregate function, while a condition in the WHERE clause may not.
A. True B. False

2.1.2 (2 points) In SQL, the WHERE clause is evaluated before the GROUP BY clause.
A. True B. False

2.2 Consider the following relational schema:

user(*uid, name, country*)
post(*pid, uid* → *user, createdAt*)
comment(*cid, pid* → *post, uid* → *user, createdAt*)
reaction(*rid, pid* → *post, uid* → *user, type*)

where the primary keys are underlined and the foreign keys are denoted by arrows.

Answer the following questions (Choose **only one** option):

2.2.1 (4 points) Which of the following queries finds all users who have written at least one post, returning *uid* and *name*?

A. SELECT *u.uid*, *u.name*
FROM *user u*
WHERE *u.uid* IN (*post*);

B. SELECT DISTINCT *u.uid*, *u.name*
FROM *user u* JOIN *post p* ON *u.uid* = *p.uid*;

C. SELECT *uid*, *name*
FROM *post*;

D. SELECT *uid*, *name*
FROM *user*
LEFT JOIN *post* ON *user.uid* = *post.uid*;

E. None of the above

2.2.2 (4 points) Which of the following queries finds each post and the total number of reactions it has received, returning *pid* and reaction count?

A. SELECT *pid*, COUNT(*)
FROM *reaction*
GROUP BY *pid*;

B. SELECT *p.pid*, COUNT(*r.rid*)
FROM *post p* JOIN *reaction r* ON *p.pid* = *r.pid*;

C. SELECT *pid*, COUNT(*rid*)
FROM *post*
GROUP BY *pid*;

D. SELECT *p.pid*, COUNT(*)
FROM *post p* LEFT JOIN *reaction r* ON *p.pid* = *r.pid*;

E. Nove of the above

2.2.3 (4 points) Which of the following queries finds all posts that have received reactions from more than 5 distinct users, returning *pid*?

A. SELECT *pid*
FROM *reaction*
WHERE COUNT(*uid*) > 5;

B. SELECT DISTINCT *pid*
FROM *reaction*
WHERE *uid* > 5;

C. SELECT *pid*
FROM *reaction*
GROUP BY *pid*
HAVING COUNT(DISTINCT *uid*) > 5;

D. SELECT *pid*
FROM *post*
GROUP BY *pid*
HAVING COUNT(*uid*) > 5;

E. Nove of the above

2.2.4 (4 points) Which of the following queries finds users who have commented on their own posts, returning *uid* and *name*?

A. SELECT DISTINCT *uid*, *name*
FROM *user*
WHERE *uid* IN (*comment*);

B. SELECT DISTINCT *u.uid*, *u.name*
FROM *user u*
JOIN *post p* ON *u.uid* = *p.uid*
JOIN *comment c* ON *p.pid* = *c.pid* AND *c.uid* = *u.uid*;

C. SELECT DISTINCT *u.uid*, *u.name*
FROM *user u*
JOIN *comment c* ON *u.uid* = *c.uid*;

D. SELECT DISTINCT *uid*, *name*
FROM *post*
JOIN *comment* ON *post.pid* = *comment.pid*;

E. Nove of the above

3. ER Modeling (16 points)

Answer the following questions (Choose **only one** option):

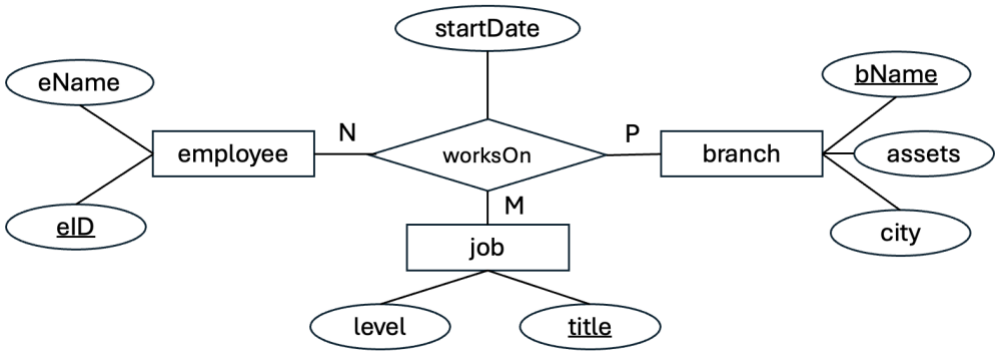
3.1 (2 points) Which of the following attributes is best modeled as a composite attribute?
- A. Age
- B. Salary
- C. Address
- D. GPA

3.2 (3 points) In Chen notation, a 1:N relationship between entity types A and B means:
- A. Each A is related to exactly one B, and each B to exactly one A
- B. Each A can be related to many B, but each B to at most one A
- C. Each A can be related to many B, and each B to many A
- D. Each A is related to exactly one B, and each B to many A

3.3 (3 points) A cardinality constraint of [0, 1] on an entity's participation in a relationship means:
- A. The entity must participate in exactly one relationship instance
- B. The entity may participate in zero or one relationship instance
- C. The entity may participate in one or more relationship instances
- D. The entity must participate in at least one relationship instance

3.4 (2 points) Which of the following is a characteristic of a weak entity?
- A. It has its own primary key
- B. It does not participate in any relationships
- C. It is identified using a partial key and a related strong entity
- D. It cannot have attributes

3.5 (6 points) Which combinations of relations should be created for the following ER diagram? Choose **one or more** options:



A. employee(eID, eName)

B. employee(eID, eName, worksOn)

C. job(title, level)

D. job(title, level, worksOn)

E. branch(bName, assets, city)

F. branch(bName, assets, city, worksOn)

G. startDate(worksOn)

H. worksOn(employee, job, branch, startDate,)

I. worksOn(startDate, employee, job, branch)

J. worksOn(eID → employee, title → job, bName → branch, startDate)

K. worksOn(eID → employee, title → job, bName → branch, startDate)

L. worksOn(eID → employee, title → job, branch, startDate)

M. worksOn(startDate, eID → employee, title → job, bName → branch)

4. Design Theory (14 points)

4.1 Evaluate whether the following statements about the design theory are true or false.

4.1.1 (2 points) If a functional dependency X → Y holds on a relation instance r, for every pair of tuples in r with the same X values, they must also have the same Y values.
A. True B. False

4.1.2 (2 points) Every functional dependency X→Y is trivial if X contains only a single attribute.
A. True B. False

4.2 Consider a relation schema $R = (A, B, C, D, E)$ with the following functional dependencies (FDs):

$$A \rightarrow BC$$
$$C \rightarrow D$$
$$DE \rightarrow A$$

Answer the following questions (Choose **one or more** options):

4.2.1 (4 points) Which of the following attributes (sets) can be the **candidate key** of R? (Write the corresponding number(s) on the answer sheet.)

(1) A	(2) C	(3) D	(4) E
(5) AC	(6) BC	(7) CD	(8) DE
(9) ABC	(10) $ABCD$	(11) $BCDE$	(12) $ABCDE$

(13) None of the above

4.2.2 (3 points) Which of the following functional dependencies violates BCNF but does not violate 3NF?

A. $A \rightarrow B$
B. $C \rightarrow D$
C. $DE \rightarrow A$
D. $A \rightarrow BC$
E. None of the above

4.2.3 (3 points) Consider decomposing R into two relations:

$$R_1(A, B, C)$$
$$R_2(A, C, D, E)$$

Which of the following statements is correct?

A. The decomposition is lossless and dependency-preserving
B. The decomposition is lossless but not dependency-preserving
C. The decomposition is dependency-preserving but not lossless
D. The decomposition is neither lossless nor dependency-preserving

5. Physical Design (12 points)

Answer the following questions (Choose **only one** option):

5.1 (2 points) In a DBMS, a block (page) is:

A. A logical record stored in memory
B. The smallest unit of data transfer between disk and memory
C. A set of tuples from different relations
D. A pointer to a disk location

5.2 (2 points) Which statement about a B+-tree is correct?

A. All records are stored in internal nodes
B. Internal nodes contain only keys and pointers
C. Leaf nodes do not contain pointers
D. The tree is not balanced

5.3 Consider the following B+ Tree with degree $d = 4$.

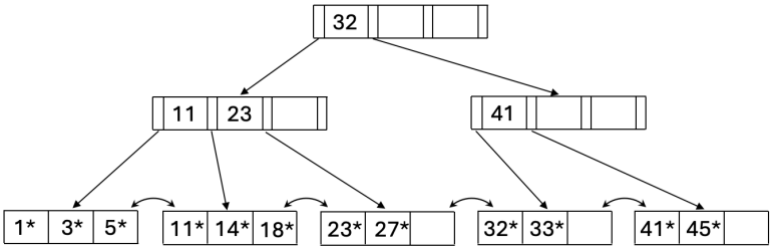


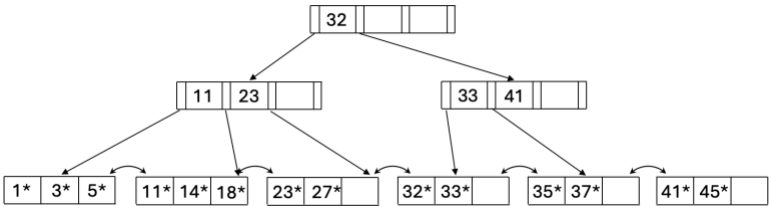
Figure 5: B+ Tree with $d = 4$

Please make the following assumptions:

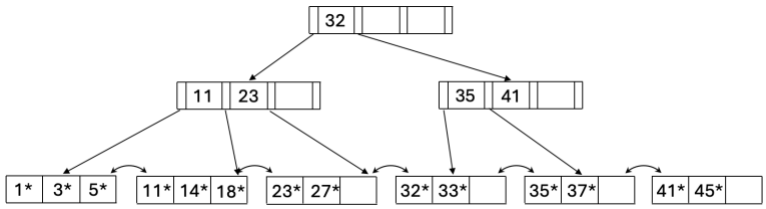
- The left pointer of a key k in a non-leaf node leads towards keys less than k, while the right pointer leads towards keys greater than or equal to k.
- A leaf node underflows when the ***number of keys*** goes below $\lceil \frac{d-1}{2} \rceil$.
- An internal node underflows when the ***number of pointers*** goes below $\lceil \frac{d}{2} \rceil$.

5.3.1 (4 points) Insert 35*, then insert 37*. What is the resulting tree?

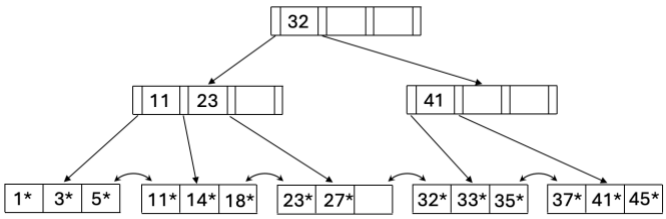
A.



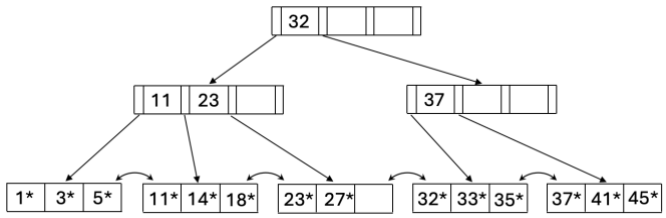
B.



C.



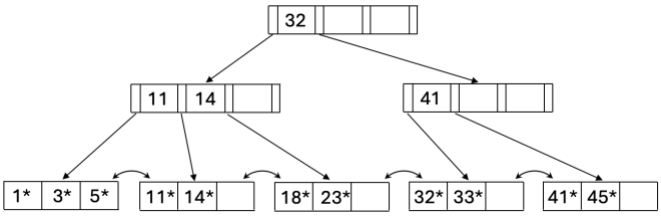
D.



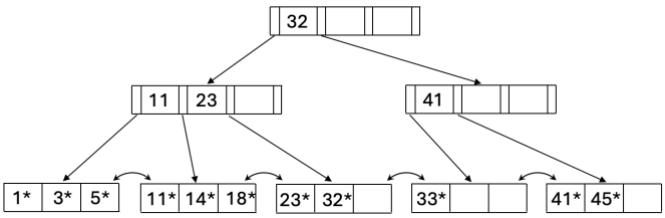
E. None of the above

5.3.2 (4 points) Consider the **original tree** in Figure 5 (ignore the operations in Question 5.3.1). Delete 27*. What is the resulting tree?

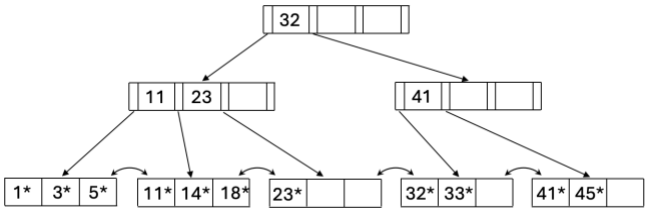
A.



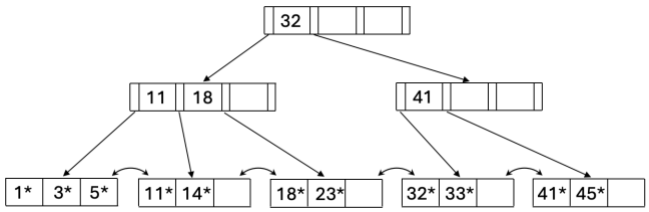
B.



C.



D.



E. None of the above

6. Query Processing and Optimization (12 points)

Answer the following questions (Choose **only one** option):

6.1 (2 points) In external sorting, what is a run?
- A. A hash bucket created during partitioning
- B. A sorted sequence of records written to disk
- C. A block of unsorted pages
- D. A join output buffer

6.2 (2 points) What is the main purpose of histograms in query optimization?
- A. To speed up query execution
- B. To store query results
- C. To estimate attribute value distributions
- D. To enforce data integrity

6.3 Relations R and S have the following properties: R has 12,000 tuples, 40 tuples per page; S has 30,000 tuples, 50 tuples per page. Assume the computer has 100 buffer pages.

6.3.1 (4 points) Assume R is the outer relation. What is the I/O cost of a simple nested-loop join:
- A. 12,000 × 30,000 + 12,000
- B. 300 × 600 + 300
- C. 12,000 × 600 + 300
- D. 300 + 600
- E. None of the above

6.3.2 (4 points) Assume that one buffer page is needed to hold the evolving output page. What is the I/O cost of a sort-merge join:
- A. 3 × (12,000 + 30,000)
- B. $2 * 300(\lceil \log_{100-1} \lceil 300/100 \rceil \rceil + 1) + 2 * 600(\lceil \log_{100-1} \lceil 600/100 \rceil \rceil + 1) + 300 + 600$
- C. $2 * 12{,}000(\lceil \log_{100-1} \lceil 12{,}000/100 \rceil \rceil + 1) + 2 * 30{,}000(\lceil \log_{100-1} \lceil 30{,}000/100 \rceil \rceil + 1) + 12{,}000 + 30{,}000$
- D. 3 × (300 + 600)
- E. None of the above

7. Transactions, Concurrency Control, and Recovery (15 points)

7.1 Evaluate whether the following statements are true or false.

7.1.1 (2 points) If all transactions follow two-phase locking (2PL), then the resulting schedule is deadlock-free.

A. True B. False

7.1.2 (2 points) Atomicity ensures that all operations of a transaction are completed, or none are applied.

A. True B. False

7.2 We consider three transactions T1, T2, and T3 consisting of the following actions:

T1: R(X) W(X) R(Y) W(Y) C
T2: R(Y) W(Z) R(X) W(X) C
T3: R(X) W(Y) R(Z) C

R(X) represents reading object X, W(X) represents writing object X, and C represents Commit. Consider the following schedule:

S:
T1: R(X) W(X) R(Y) W(Y) C
T2: R(Y) W(Z) R(X) W(X) C
T3: R(X) W(Y) R(Z) C

Answer the following questions (Choose **one or more** options):

7.2.1 (2 points) Which of the following types of conflicts can affect serializability?

A. R(X)-R(X)
B. R(X)-W(Y)
C. W(X)-R(X)
D. W(X)-W(X)

7.2.2 (3 points) Which type(s) of schedule does Schedule S belong to?

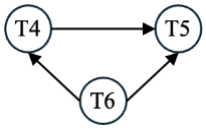
A. Serial schedule
B. Serializable schedule
C. Conflict serializable schedule
D. None of the above

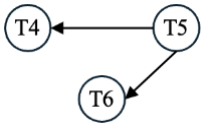
7.3 Consider three transactions T4, T5, and T6 with the following lock requests (Lock-X = exclusive lock, Lock-S = shared lock):

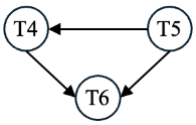
(1) T4 requests Lock-X(A)
(2) T5 requests Lock-X(B)
(3) T6 requests Lock-S(A)
(4) T4 requests Lock-X(B)
(5) T5 requests Lock-X(C)
(6) T6 requests Lock-X(C)

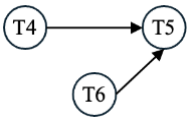
Answer the following questions (Choose **one or more** options):

7.3.1 (3 points) Which wait-for graph is correct?

A. 

C. 

B. 

D. 

7.3.2 (3 points) Choose correct statement(s)?

A. No deadlock exists
B. A deadlock exists
C. Transaction T6 is waiting for both T4 and T5 to release locks
D. None of the above